

CBSE 2027 Physics — Predicted Paper (Set 9 of 10)

SECTION A: Objective Type Questions (16 Marks)

This section consists of 12 MCQs and 4 Assertion-Reasoning questions of 1 mark each.

Q1. A point charge q is placed exactly at one of the corners of a hollow cube. The total electric flux passing through the entire cube is: (a) q/ϵ_0 (b) $q/2\epsilon_0$ (c) $q/6\epsilon_0$ (d) $q/8\epsilon_0$ (Chapter: Electric Charges and Fields) | **[Confidence: Very High]**

Q2. The electrical resistance of the depletion layer in a $p - n$ junction diode is very large because: (a) it has a high concentration of majority carriers. (b) it contains no mobile charge carriers. (c) holes and electrons are moving at the speed of light. (d) the electric field inside it is zero. (Chapter: Semiconductor Electronics) | **[Confidence: Very High]**

Q3. The magnetic field inside a long, tightly wound, straight solenoid carrying a steady current is: (a) zero. (b) uniform and strong everywhere along its axis. (c) maximum at the ends and zero at the center. (d) increasing gradually from the axis to the surface. (Chapter: Moving Charges and Magnetism) | **[Confidence: High]**

Q4. When a small magnetic needle is placed in a non-uniform magnetic field, it will experience: (a) a force and a torque. (b) a force but not a torque. (c) a torque but not a force. (d) neither a force nor a torque. (Chapter: Magnetism and Matter) | **[Confidence: High]**

Q5. The SI unit of self-inductance (Henry) is equivalent to: (a) Volt $\cdot$ second/Ampere (b) Ampere $\cdot$ second/Volt (c) Volt $\cdot$ Ampere/second (d) Joule $\cdot$ Ampere/second (Chapter: Electromagnetic Induction) | **[Confidence: Very High]**

Q6. The phase difference between the alternating current and the applied alternating EMF in a circuit is exactly $\pi/2$. Which of the following cannot be the sole constituent(s) of this circuit? (a) A pure inductor (L) alone (b) A pure capacitor (C) alone (c) An inductor (L) and a capacitor (C) in series (d) A resistor (R) and an inductor (L) in series (Chapter: Alternating Current) | **[Confidence: High]**

Q7. Which of the following electromagnetic radiations possesses the minimum wavelength? (a) Blue light (b) Gamma rays (c) Infrared rays (d) Microwaves (Chapter: Electromagnetic Waves) | **[Confidence: Very High]**

Q8. When a monochromatic beam of light travels from a rarer to a denser medium, its speed decreases. What happens to the energy carried by a single photon of this light wave? (a) It increases. (b) It decreases. (c) It remains strictly unchanged. (d) It depends on the intensity of the incident light. (Chapter: Ray Optics / Dual Nature of Radiation and Matter) | **[Confidence: Very High]**

Q9. Two coherent light beams of intensities I and $4I$ are superposed in an interference experiment. The maximum and minimum possible intensities in the resulting pattern are

respectively: (a) $5I$ and $3I$ (b) $9I$ and I (c) $5I$ and I (d) $9I$ and $3I$ (Chapter: Wave Optics) | **[Confidence: Very High]**

Q10. The work function of a metal surface is denoted by $W = h\nu_0$. Light of frequency ν falls on this metal. Photoelectric emission will occur only if: (a) $\nu < \nu_0$ (b) $\nu \geq \nu_0$ (c) $\nu = \nu_0/2$ (d) $\nu \leq \nu_0/2$ (Chapter: Dual Nature of Radiation & Matter) | **[Confidence: Very High]**

Q11. According to Bohr's model of the hydrogen atom, the angular momentum of an electron revolving in the second stationary orbit ($n = 2$) is: (a) h/π (b) $2h/\pi$ (c) $h/2\pi$ (d) $h/4\pi$ (Chapter: Atoms) | **[Confidence: High]**

Q12. The strong nuclear force between two adjacent nucleons is known to be: (a) strictly central and purely electrostatic. (b) long-range and inverse-square in nature. (c) spin-dependent and non-central. (d) heavily dependent on the charges of the nucleons. (Chapter: Nuclei) | **[Confidence: Very High]**

Q13. Assertion (A): The surface of a charged spherical conductor in electrostatic equilibrium is an equipotential surface. **Reason (R):** A conductor allows the free movement of charges to instantly equalize the potential across its entire surface, making the tangential electric field zero. (Chapter: Electrostatic Potential and Capacitance) | **[Confidence: Very High]**

Q14. Assertion (A): A proton and an alpha particle entering a uniform magnetic field perpendicularly with the exact same speed will describe circular paths of different radii.

Reason (R): The radius of the circular path of a charged particle in a magnetic field is directly proportional to its mass-to-charge ratio (m/q) for a given speed. (Chapter: Moving Charges and Magnetism) | **[Confidence: High]**

Q15. Assertion (A): When a bar magnet falls vertically through a long, hollow metallic pipe, its acceleration is strictly less than the acceleration due to gravity (g). **Reason (R):** The falling magnet induces eddy currents in the metallic pipe, which, according to Lenz's law, produce a magnetic field that opposes the motion of the magnet. (Chapter: Electromagnetic Induction) | **[Confidence: Very High]**

Q16. Assertion (A): A p-n junction diode effectively acts as a one-way switch in an electrical circuit. **Reason (R):** It offers a very low resistance when forward-biased and a heavily high resistance when reverse-biased. (Chapter: Semiconductor Electronics) | **[Confidence: Very High]**

SECTION B: Very Short Answer Type Questions (10 Marks)

This section consists of 5 questions of 2 marks each.

Q17. Calculate the exact magnitude of the electric field strength required to just support a tiny water drop of mass 10^{-7} kg carrying a charge of 1.6×10^{-19} C from falling under gravity. (Take $g = 9.8 \text{ m/s}^2$). (Chapter: Electric Charges and Fields) | **[Confidence: Very High]**

Q18. Draw a typical V-I graph for a semiconductor device like Gallium Arsenide (GaAs). On your graph, clearly mark the non-linear region and the region where it exhibits negative

resistance. (*Chapter: Current Electricity*) | **[Confidence: High]**

Q19. Define the magnetic susceptibility of a material. Based on their susceptibility values, classify Bismuth and Aluminum into their respective magnetic categories. (*Chapter: Magnetism and Matter*) | **[Confidence: Very High]**

Q20. What is a choke coil? State one explicit reason why a choke coil is vastly preferred over a standard resistor to control alternating current (AC) in a circuit. (*Chapter: Alternating Current*) | **[Confidence: High]**

Q21. Define the term 'mass defect' of a nucleus. A nucleus has an actual mass M , atomic number Z , and mass number A . Write the mathematical expression for its mass defect in terms of the rest masses of a free proton (m_p) and a free neutron (m_n). (*Chapter: Nuclei*) | **[Confidence: Very High]**

SECTION C: Short Answer Type Questions (21 Marks)

This section consists of 7 questions of 3 marks each.

Q22. Two point charges q and $-3q$ are placed fixed on the x-axis, separated by a distance d . Determine the exact location where a third point charge $2q$ should be placed such that it will not experience any net electrostatic force. (*Chapter: Electric Charges and Fields*) | **[Confidence: High]**

Q23. Describe the trajectory of a charged particle of mass m and charge q moving in a uniform magnetic field $\vec{B}$ when its initial velocity $\vec{v}$ enters the field making an angle θ (where $0^\circ < \theta < 90^\circ$). Deduce the mathematical expression for the pitch of its helical path. (*Chapter: Moving Charges and Magnetism*) | **[Confidence: Very High]**

Q24. State Lenz's Law of electromagnetic induction. A rectangular metallic loop and a circular metallic loop are pulled out of a uniform magnetic field region into a field-free region with a constant velocity v . In which of the two loops do you expect the induced EMF to remain constant during the entire passage out of the field region? Justify your answer. (*Chapter: Electromagnetic Induction*) | **[Confidence: Very High]**

Q25. What is 'displacement current'? Briefly explain the logical inconsistency in Ampere's circuital law that led James Clerk Maxwell to introduce this concept. Write down the complete, modified Ampere-Maxwell circuital law. (*Chapter: Electromagnetic Waves*) | **[Confidence: Very High]**

Q26. What is meant by the diffraction of light? Draw a neat graph showing the variation of intensity with angle θ in a single-slit diffraction experiment. Write one fundamental feature that strictly distinguishes a diffraction pattern from an interference pattern. (*Chapter: Wave Optics*) | **[Confidence: Very High]**

Q27. Using Bohr's second postulate of the quantization of angular momentum ($L = nh/2\pi$), rigorously show that the circumference of the electron's orbit in the n^{th} stationary state of a

hydrogen atom is exactly equal to n times the de-Broglie wavelength (λ) associated with the revolving electron. (*Chapter: Atoms / Dual Nature*) | **[Confidence: High]**

Q28. Draw the standard V-I characteristics of a p-n junction diode in both forward and reverse bias on the same graph. Clearly define the terms "cut-in voltage" (knee voltage) and "reverse breakdown voltage" from your graph. (*Chapter: Semiconductor Electronics*) | **[Confidence: Very High]**

SECTION D: Case-Study Based Questions (8 Marks)

This section consists of 2 questions of 4 marks each. Questions usually have internal sub-parts.

Q29. Case Study 1: Temperature Dependence of Resistivity *Passage:* The electrical resistivity of a material is highly dependent on its temperature. For pure metals, resistivity increases with an increase in temperature because the collision frequency of electrons with lattice ions increases, reducing the relaxation time. Conversely, for semiconductors, resistivity drops exponentially as temperature rises due to the breaking of covalent bonds, which exponentially increases the number density of charge carriers. Interestingly, certain alloys like nichrome and manganin show very weak temperature dependence. (a) Write the empirical formula showing the variation of the resistivity (ρ) of a metal with a small change in temperature (ΔT). (1 mark) (b) Why are alloys like constantan or manganin specifically chosen to construct standard wire-bound resistors? (1 mark) (c) A silver wire has an electrical resistance of $2.1\ \Omega$ at 27.5°C , and a resistance of $2.7\ \Omega$ at 100°C . Determine the temperature coefficient of resistivity (α) of silver. **OR** Explain qualitatively, using the expression $\rho = \frac{m}{ne^2\tau}$, why the resistivity of a typical intrinsic semiconductor tightly decreases with a rise in temperature. (2 marks) (*Chapter: Current Electricity*) | **[Confidence: Very High]**

Q30. Case Study 2: Total Internal Reflection (TIR) *Passage:* When light travels from an optically denser medium to an optically rarer medium, it bends away from the normal. If the angle of incidence is gradually increased, the angle of refraction also increases until, at a specific angle of incidence called the critical angle, the refracted ray grazes the interface. If the angle of incidence exceeds this critical angle, the light is completely reflected back into the denser medium. This phenomenon, known as Total Internal Reflection (TIR), is the operational basis for optical fibers, diamond brilliance, and mirages. (a) State the mathematical relationship between the refractive index (μ) of a denser medium (with respect to air) and its critical angle (i_c). (1 mark) (b) Does the critical angle for a given pair of media depend on the color (wavelength) of the incident light? Briefly state why. (1 mark) (c) An optical fiber consists of a core of refractive index 1.50 and a cladding of refractive index 1.44 . Calculate the critical angle for the core-cladding interface. (You may leave the answer in the form of $\sin^{-1}$). **OR** Draw a neat ray diagram to demonstrate how a right-angled isosceles glass prism can be used to deviate a light ray by exactly 90° using TIR. (2 marks) (*Chapter: Ray Optics and Optical Instruments*) | **[Confidence: Very High]**

SECTION E: Long Answer Type Questions (15 Marks)

This section consists of 3 questions of 5 marks each. All questions have an internal "OR" choice.

Q31. (a) Draw a labelled diagram of a parallel plate capacitor. Derive the mathematical expression for the capacitance of a parallel plate capacitor with plate area A and separation d , when the entire space between the plates is filled with a dielectric medium of dielectric constant K . **(b)** A parallel plate capacitor with air between the plates has a measured capacitance of 8 pF . What will be its new capacitance if the distance between the plates is reduced by half, and the space between them is completely filled with a substance of dielectric constant $K = 6$? **OR (a)** State Gauss's Law in electrostatics. **(b)** Use Gauss's Law to derive an expression for the electric field intensity at a point near a uniformly charged, infinite, thin plane sheet with surface charge density σ . **(c)** Two large, thin metal plates are held parallel and close to each other. On their inner faces, the plates have uniform surface charge densities of opposite signs and of magnitude $17.0 \times 10^{-22} \text{ C/m}^2$. Calculate the electric field $\vec{E}$: (i) in the outer region of the first plate, and (ii) in the inner region between the two plates. (Chapter: Electrostatic Potential and Capacitance / Electric Charges and Fields) | **[Confidence: Very High]**

Q32. (a) Draw a neat labelled diagram of a Moving Coil Galvanometer. State its underlying working principle. **(b)** Derive an expression for the torque experienced by the rectangular current-carrying coil when placed in a uniform magnetic field. **(c)** What is the specific purpose of employing a radial magnetic field in the galvanometer? How is this radial field physically produced? **(d)** Define the current sensitivity of the galvanometer. **OR (a)** An alternating voltage $V = V_0 \sin(\omega t)$ is applied across a series combination of a resistor R , an inductor L , and a capacitor C . Draw the corresponding phasor diagram. **(b)** Using the phasor diagram, logically derive the expression for the total impedance (Z) of the circuit and the phase angle (ϕ) by which the current leads or lags the voltage. **(c)** State the condition under which electrical resonance occurs in this series LCR circuit. Obtain the exact expression for the resonant frequency (ν_r). (Chapter: Moving Charges and Magnetism / Alternating Current) | **[Confidence: Very High]**

Q33. (a) State the fundamental postulates of Huygens' wave theory of light. **(b)** Using Huygens' principle, construct a diagram to show the reflection of a plane wavefront incident on a plane reflecting surface. **(c)** Hence, logically deduce the laws of reflection (angle of incidence = angle of reflection) from your geometrical construction. **OR (a)** Consider a point object placed on the principal axis of a convex spherical refracting surface (of radius of curvature R) which separates a rarer medium (refractive index n_1) from a denser medium (refractive index n_2). Draw the ray diagram and derive the relation: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$. **(b)** An air bubble is formed inside a solid glass sphere of radius 10 cm (refractive index of glass = 1.5). The bubble is situated at a distance of 4 cm from the center of the sphere. Determine the apparent position of the bubble when viewed directly from the outside through the nearest surface. (Chapter: Wave Optics / Ray Optics and Optical Instruments) | **[Confidence: Very High]**

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